

SIMATS ENGINEERING
Department of Computer Science and Engineering

COURSE NAME:	Cloud Computing and Big Data Analytics – Model Practical Examination	COURSE CODE:	CSA1522
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Register Number	192511377

MODEL PRACTICAL EXAMINATION – SET 6

Answer all questions. Each question carries equal marks (20).

Total marks: 100

1. Create a simple cloud software application for Online Super Market using any Cloud Service Provider to demonstrate SaaS, with the template of Customer Name, Address, Phone, Email, Items, Quantity, etc. along with the item classification form.
2. Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Name of the Student, Register No., Subjects, Marks, Average, Percentage, Rank, Overall Performance, etc.
3. Install VirtualBox with different flavours of Linux or Windows OS on top of Windows 10 using custom installation.
4. Write a procedure to run virtual machines of different configurations and check how many virtual machines can be utilized at a particular time.

1. Create a simple cloud software application for Online Super Market using any Cloud Service Provider to demonstrate SaaS, with the template of Customer Name, Address, Phone, Email, Items, Quantity, etc. along with the item classification form.

Aim:

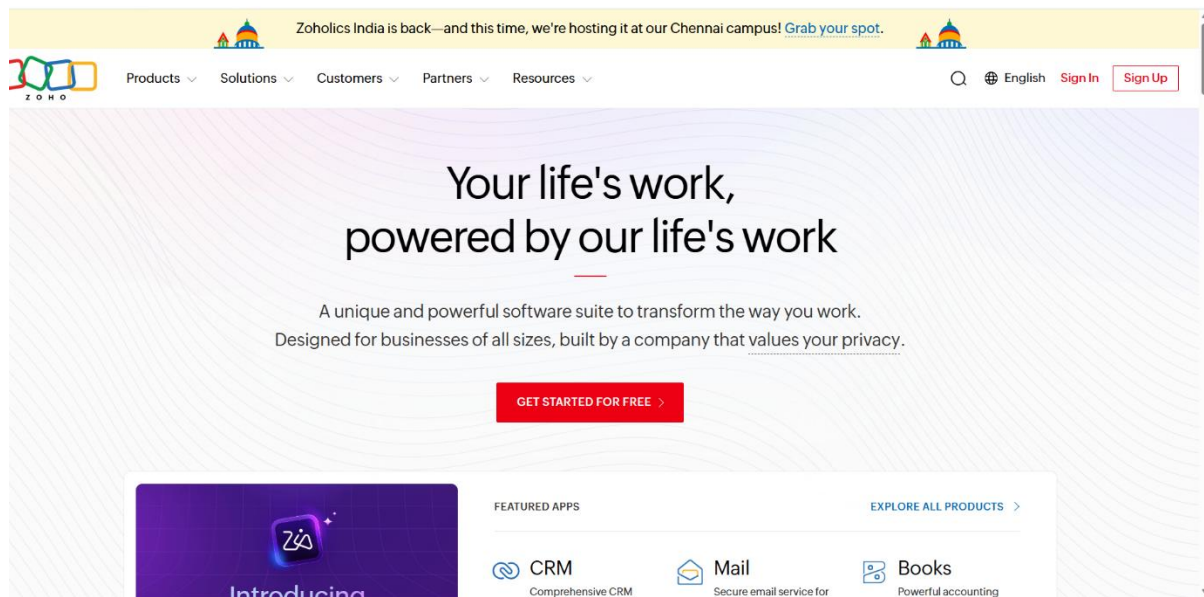
To create a cloud-based Online Supermarket System using Zoho Creator.

Steps:

1. Create a new application.
2. Create an **Online Supermarket** form.
3. Add fields: Customer Name, Address, Phone Number, Email, Item Name, Quantity, Price, Item Classification, and Total Amount.
4. Create reports to view customer orders.
5. Enter sample records and verify the application.

Implemented:

Step 1: Go to zoho.com



STEP 2: Login to your Zoho account and open Zoho Creator

 Desk Helpdesk software to deliver great customer support. TRY NOW >	 Payroll Effortless payroll processing software for businesses. TRY NOW >	 Recruit Intuitive recruiting platform built to provide hiring solutions. TRY NOW >	 Projects Manage, track, and collaborate on projects with teams. TRY NOW >
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Apps you've accessed

 Cliq	 Creator	 Forms	 People
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STEP 3: Click Create New Application and select Create from Scratch

Create New Application

	Create using Zia Build a functional application effortlessly using GenAI Create	
 Create from scratch Create a new application for your business needs Create	 Create from gallery Pick a pre-built application that suits your requirement Pick	 Import from file Import your existing data to create an application Import

STEP 4: Enter Online Supermarket System as the application name and click Create Application.

Create from scratch

Application Name *

Online Supermarket System

☐ Enable Environment [Learn More](#)

Create Application

Step 5: Add fields: Customer Name, Address, Phone Number, Email, Item Name, Quantity, Price, Item Classification, and Total Amount.

Online Supermarket Management

Online Supermarket Mana...

Products

Dashboard

Order Items

Products

+ Products

Products Report

Customer Address...

Item Classificatio...

Orders

Customers

Products

Name

Brand

Classification

-Select-

Price

#####.##

Stock Quantity

#####

Created At

dd-MMM-yyyy HH:mm:ss

Updated At

dd-MMM-yyyy HH:mm:ss

Submit

Reset

Online Supermarket Management

Online Supermarket Mana...

Dashboard

Order Items >

Products >

Customer Address... >

+ Customer Address...

Customer Address...

Item Classificatio... >

Orders >

Customers >

Customer Addresses

Customer

Address

Address Line 1

☐ Is Primary

Created At

Updated At

Submit

Reset

STEP 6: Save the form Online Supermarket System, enter sample booking records, and view them in the System

Online Supermarket Management

Online Supermarket Mana...

Dashboard

Order Items >

Products >

Customer Address... >

+ Customer Address...

Customer Addresses Report

Item Classificatio... >

Orders >

Customers >

Customer Addresses Report

Customer

Address

Is Primary

Created At

Updated At

1	654 Cedar Blvd	true	19-Jul-2026 16:20:00	19-Jul-2026 16:20:00
2				
3				
1	321 Elm Street	false	18-Jul-2026 14:45:00	18-Jul-2026 14:45:00
2				
3				
1	789 Pine Road	true	17-Jul-2026 11:00:00	17-Jul-2026 11:00:00
2				
3				
1	456 Oak Lane	false	16-Jul-2026 10:15:00	16-Jul-2026 10:15:00
2				
3				
1	123 Maple Ave	true	15-Jul-2026 09:30:00	15-Jul-2026 09:30:00
2				
3				

Online Supermarket Management

Online Supermarket Mana...

Dashboard

Order Items >

Products >

Products >

Products Report

Customer Address... >

Item Classificatio... >

Orders >

Products Report

Product ID

Name

Brand

Classification

Price

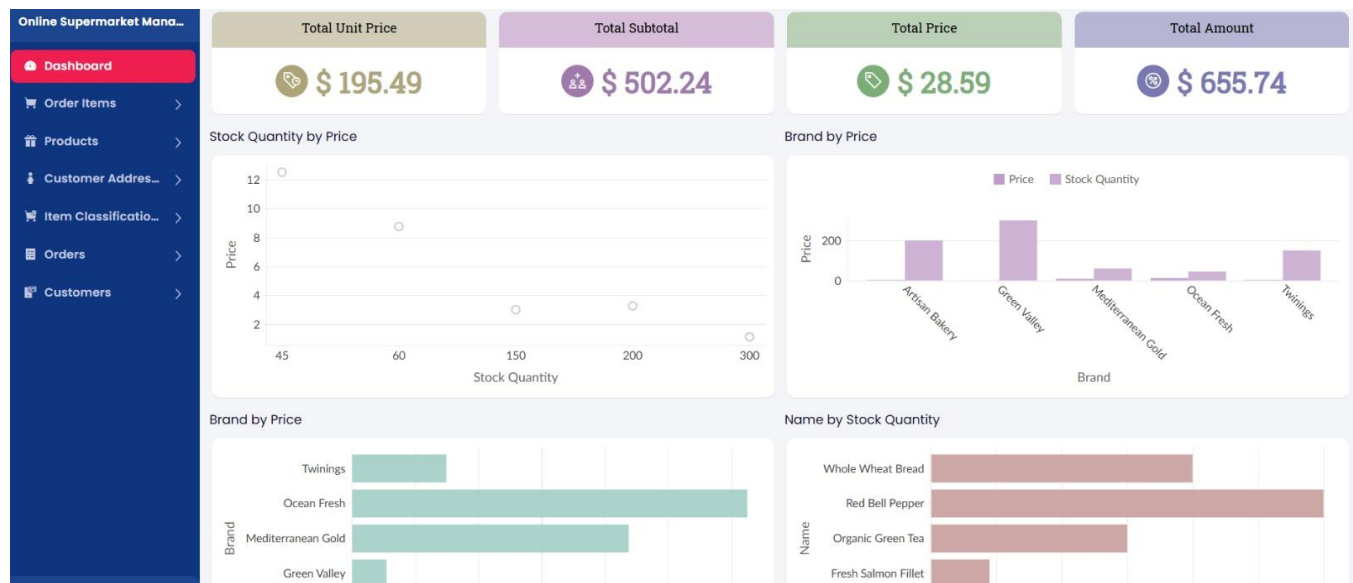
Stock Quantity

Created At

Updated At

5	Extra Virgin Olive Oil	Mediterranean Gold	1	\$ 8.75	60	19-Jul-2026 11:30:00	31-Jul-2026 12:30:00
4	Red Bell Pepper	Green Valley	1	\$ 1.10	300	18-Jul-2026 10:00:00	31-Jul-2026 11:00:00
			2				
			3				
3	Whole Wheat Bread	Artisan Bakery	1	\$ 3.25	200	17-Jul-2026 07:30:00	31-Jul-2026 08:20:00
2	Fresh Salmon Fillet	Ocean Fresh	1	\$ 12.50	45	16-Jul-2026 08:00:00	31-Jul-2026 09:45:00
			2				
			3				
1	Organic Green Tea	Twinnings	1	\$ 2.99	150	15-Jul-2026 09:30:00	31-Jul-2026 10:15:00
			2				

Step 7: The Software has Been Created



Result: The Online Super Market cloud application was successfully created with customer details, item classification, quantity, and billing-related fields. The application demonstrated the basic working of SaaS on a cloud platform.

2.. Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Name of the Student, Register No., Subjects, Marks, Average, Percentage, Rank, Overall Performance, etc

Aim:

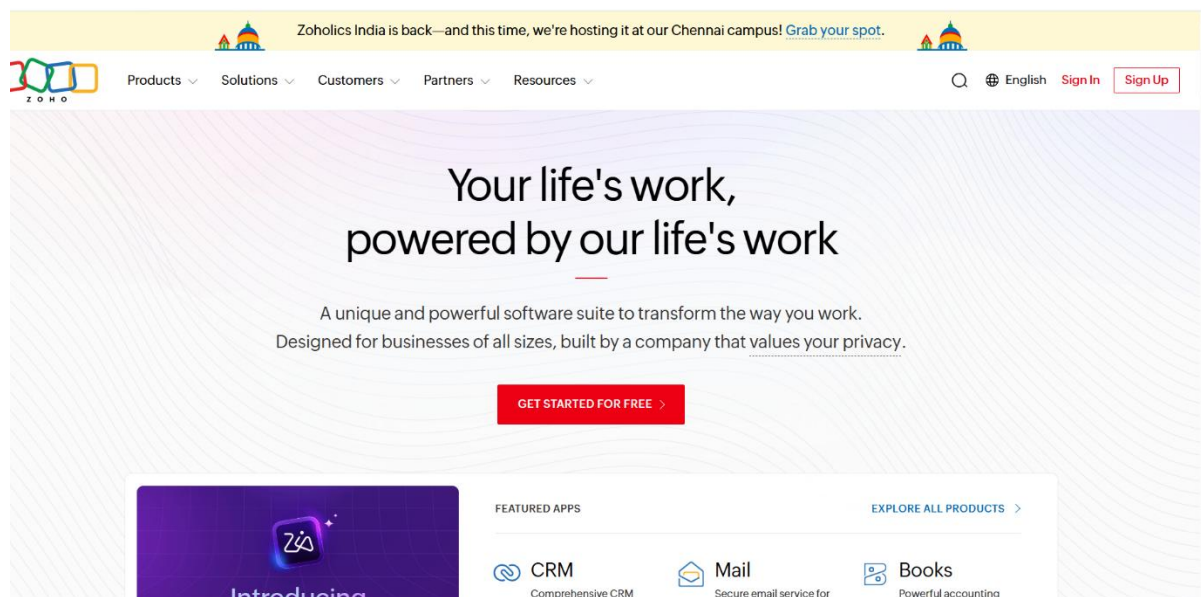
To develop a cloud-based Mark Sheet Processing System using Zoho Creator.

Steps:

1. Create a new application.
2. Create a Student Mark Sheet form.
3. Add fields: Student Name, Register Number, Subject Marks, Total, Average, Percentage, and Rank.
4. Use formulas to calculate Total, Average, and Percentage.
5. Save the records and verify the generated mark sheet.

Implemented

Step 1: Go to zoho.com



STEP 2: Login to your Zoho account and open Zoho Creator



Desk
Helpdesk software to deliver great customer support.
[TRY NOW >](#)



Payroll
Effortless payroll processing software for businesses.
[TRY NOW >](#)



Recruit
Intuitive recruiting platform built to provide hiring solutions.
[TRY NOW >](#)



Projects
Manage, track, and collaborate on projects with teams.
[TRY NOW >](#)

Apps you've accessed



Cliq



Creator



Forms



People

STEP 3: Click Create New Application and select Create from Scratch

Create New Application



Create using Zia
Build a functional application effortlessly using GenAI
[Create](#)



Create from scratch
Create a new application for your business needs
[Create](#)



Create from gallery
Pick a pre-built application that suits your requirement
[Pick](#)



Import from file
Import your existing data to create an application
[Import](#)

STEP 4: Enter Mark Sheet Processing System as the application name and click Create Application.

☒

☐

☐

☐

Create from scratch

Application Name *

: Mark Sheet Processing System

☐ Enable Environment [Learn More](#)

Create Application

Step 5: Add fields: Student Name, Register Number, Subject Marks, Total, Average, Percentage, and Rank.

Mark Sheet Processing Sys...

Dashboard

Mark Entry

+ Mark Entry

Mark Entry Report

Board

Performance Sum...

Student

Subject

Mark Entry

Student

Subject

Exam Type

Obtained Marks

Date Recorded

Grade

Remarks

Submit

Reset

Mark Sheet Processing Sys...

Dashboard

Mark Entry

+ Performance Sum...

Performance Sum...

Board

Student

Subject

Performance Summary

Student

Academic Year

Total Marks

Average Marks

Percentage

Rank

Overall Performance

Status

Submit

Reset

STEP 6: Save the form, enter sample booking records, and view them in the Mark Sheet Processing System.

Mark Sheet Processing Sys...

Dashboard

Mark Entry

+ Mark Entry

Mark Entry Report

Board

Performance Sum...

Student

Subject

Mark Entry Report

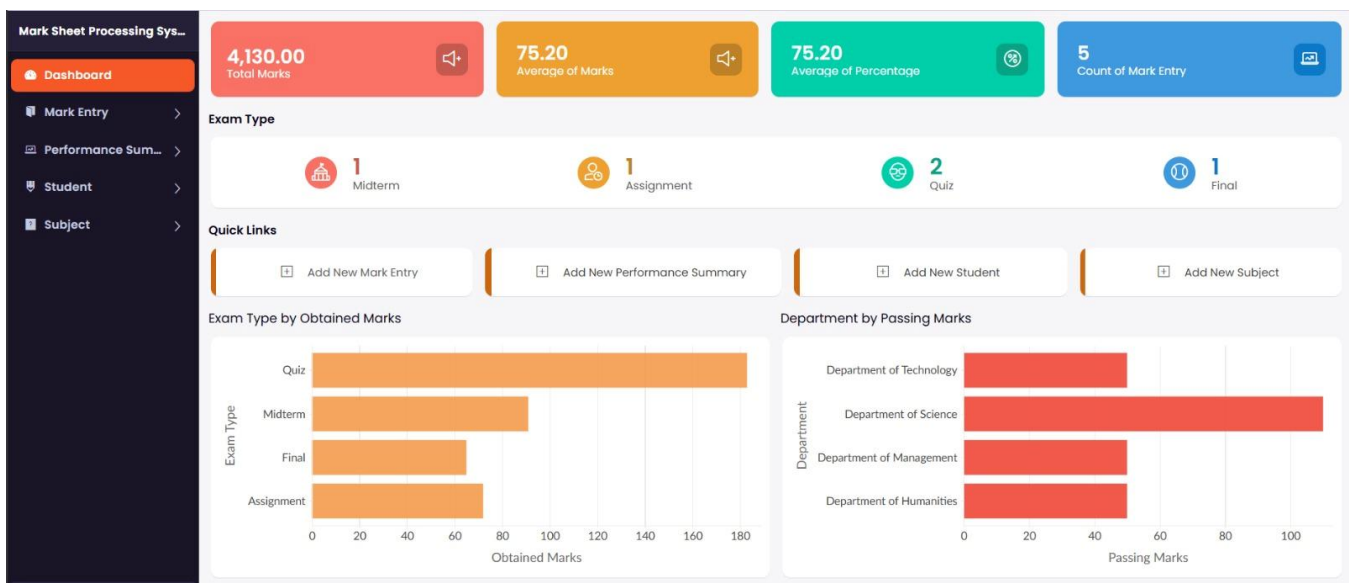
Student	Subject	Exam Type	Obtained Marks	Date Recorded	Grade	Remarks
STU2026001 STU2026003	MATH301 CS101 BUS305 ART101	Midterm	91	31-Jul-2026	C+	Outstanding work demonstrated throughout the term
STU2026001 STU2026002 STU2026003 STU2026004 STU2026005	MATH301 CS101	Final	65	31-Jul-2026	C-	Average results across the board
STU2026001	MATH301 CS101	Assignment	72	31-Jul-2026	A	Needs more practice in mathematics
STU2026001 STU2026002	MATH301	Quiz	88	31-Jul-2026	A-	Good effort with room for improvement
STU2026001 STU2026002 STU2026003 STU2026004	CS101	Quiz	95	31-Jul-2026	B	Excellent performance in all subjects

Mark Sheet Processing System

Performance Summary Report

Student	Academic Year	Total Marks	Average Marks	Percentage	Rank	Overall Performance	Status
STU2026001 STU2026002	2024-2025	710	64.00	64.00	5	Needs Improvement	Completed
STU2026001 STU2026002	2024-2025	760	69.00	69.00	4	Satisfactory	Completed
STU2026001 STU2026003	2024-2025	820	74.50	74.50	3	Satisfactory	Completed
STU2026001	2024-2025	890	81.00	81.00	2	Fail	Active
STU2026001 STU2026002 STU2026003 STU2026004 STU2026005	2024-2025	950	87.50	87.50	1	Outstanding	Active

Step 7: The Software has Been Created:



Result: The Mark Sheet Processing System was successfully created with student details, subject marks, average, percentage, rank, and overall performance. The application demonstrated SaaS-based online mark processing.

3. Install VirtualBox with different flavours of Linux or Windows OS on top of Windows 10 using custom installation.

Aim

To install VirtualBox and create a virtual machine with a different Linux or Windows operating system on the existing Windows environment.

Short Procedure

1. Download and install Oracle VirtualBox on the Windows system.
2. Download the ISO file of the required Linux/Windows OS.
3. Open VirtualBox and select Create New Virtual Machine.
4. Enter the VM name and select the downloaded ISO file.
5. Allocate suitable RAM, CPU cores, and virtual hard-disk space.
6. Start the virtual machine and follow the OS installation instructions.
7. Complete the installation and restart the virtual machine.
8. Verify that the newly installed OS runs successfully inside the virtual environment.

FIG : CREATING VM AND PROVIDING ISO FILE

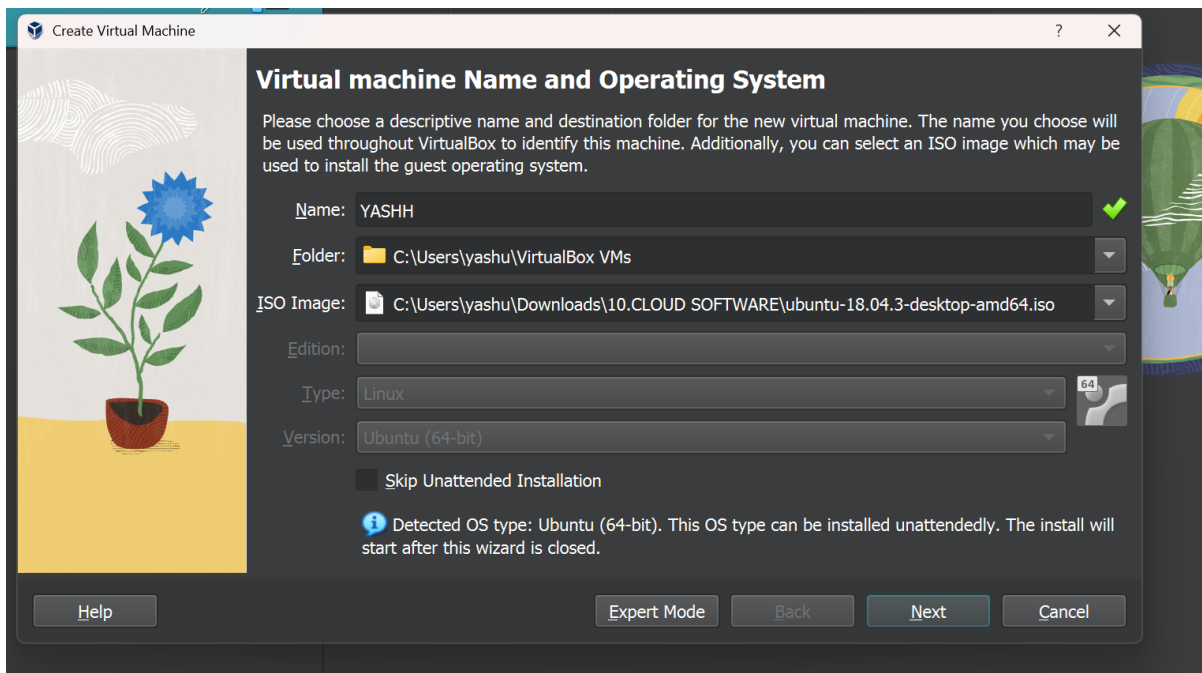


FIG :PROVIDING CREDENTIALS FOR THE VM

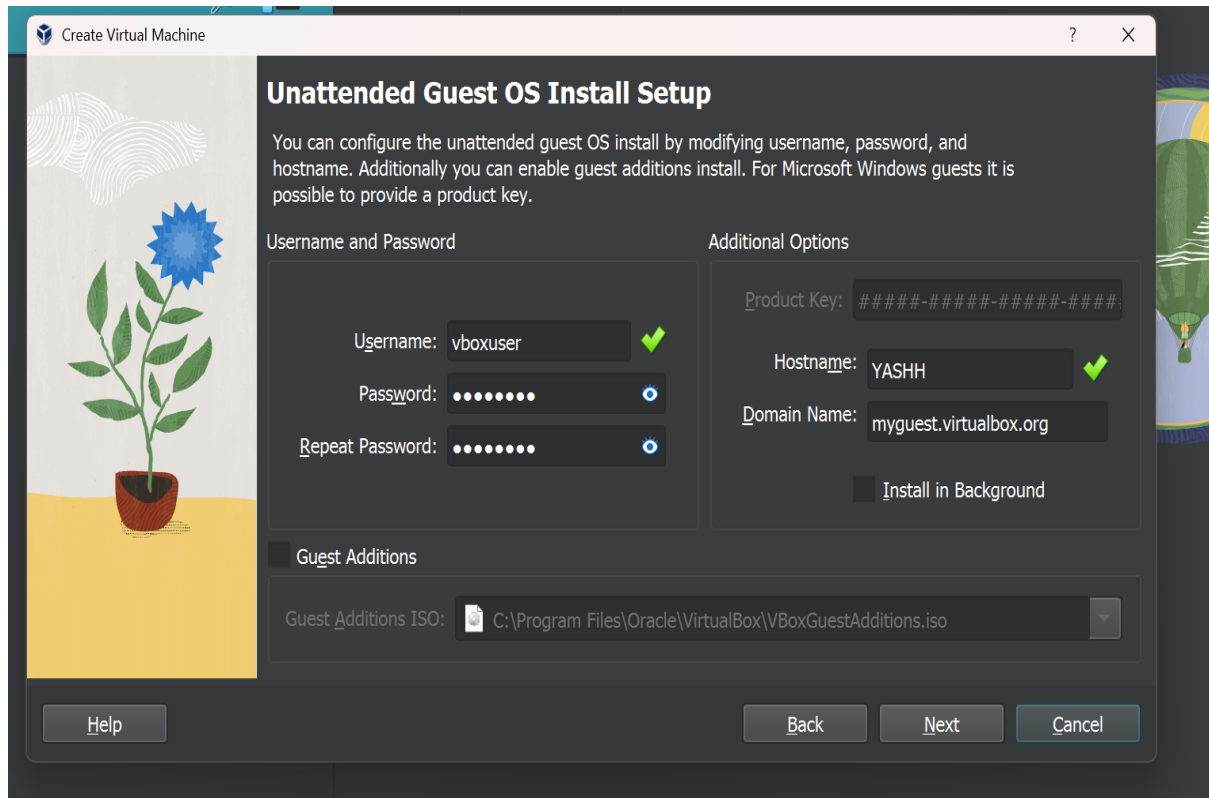


FIG : CREATING VIRTUAL HARD DISK WITH REQUIRED CONFIGURATION

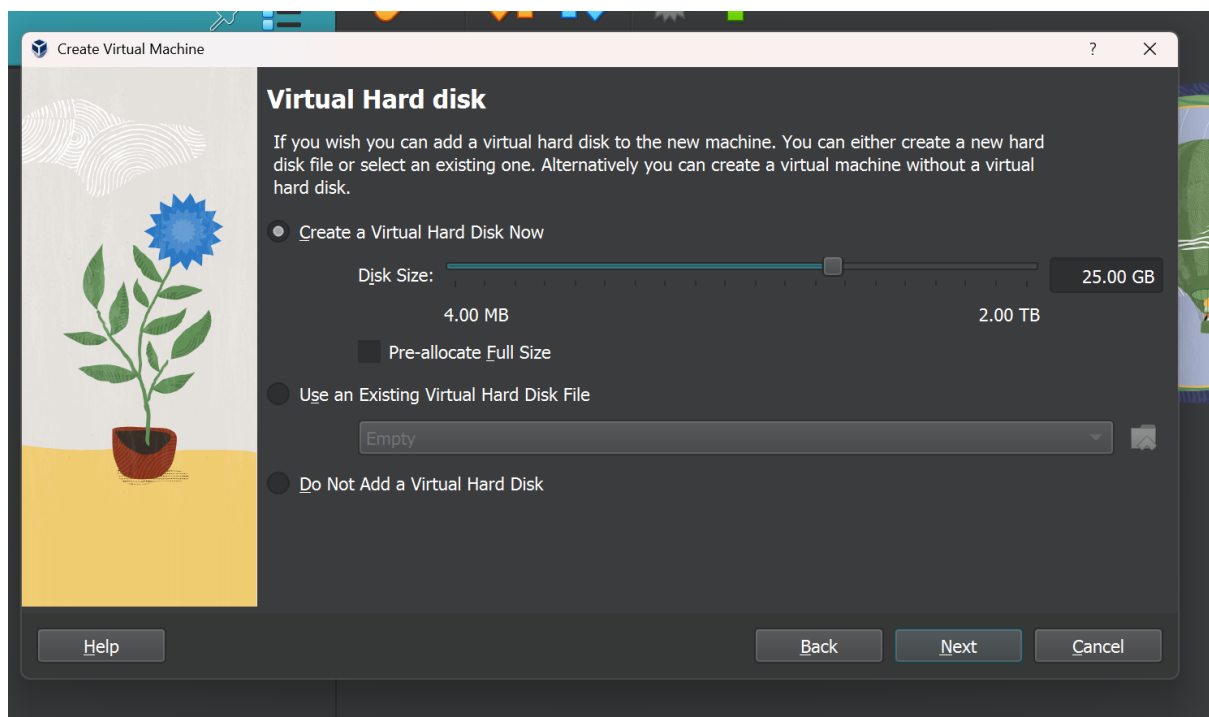


FIG : CONFIGURING THE VM

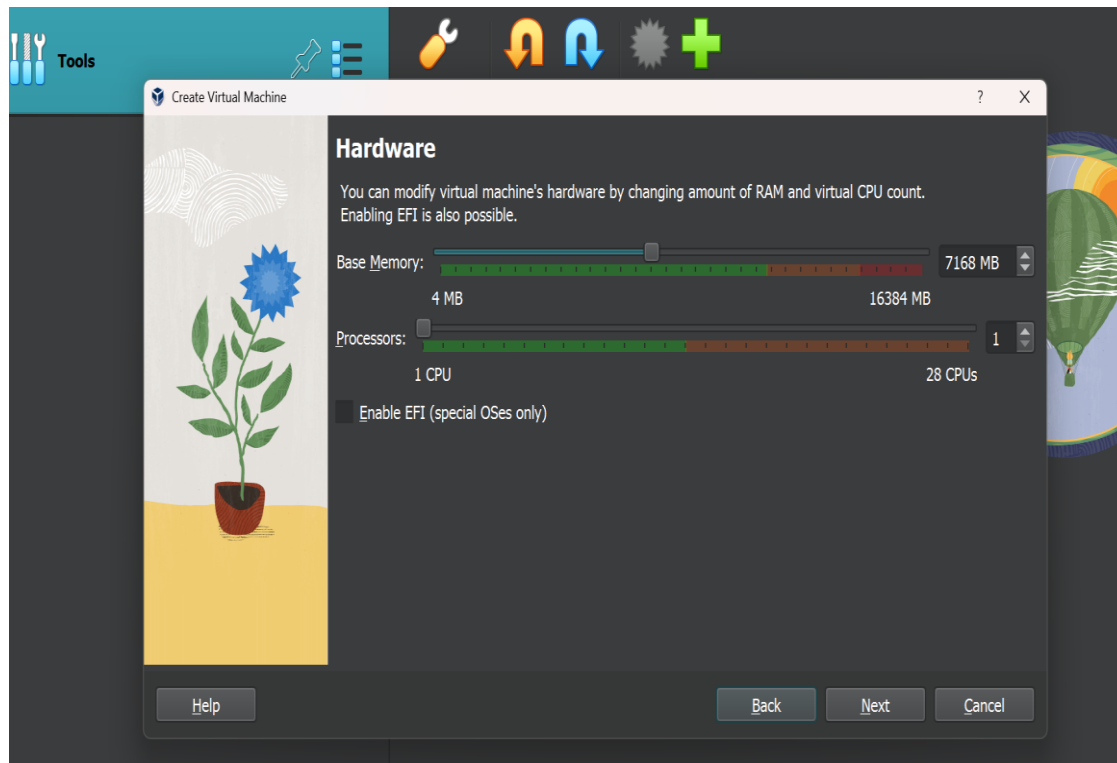


FIG : LAST STEP OF VM

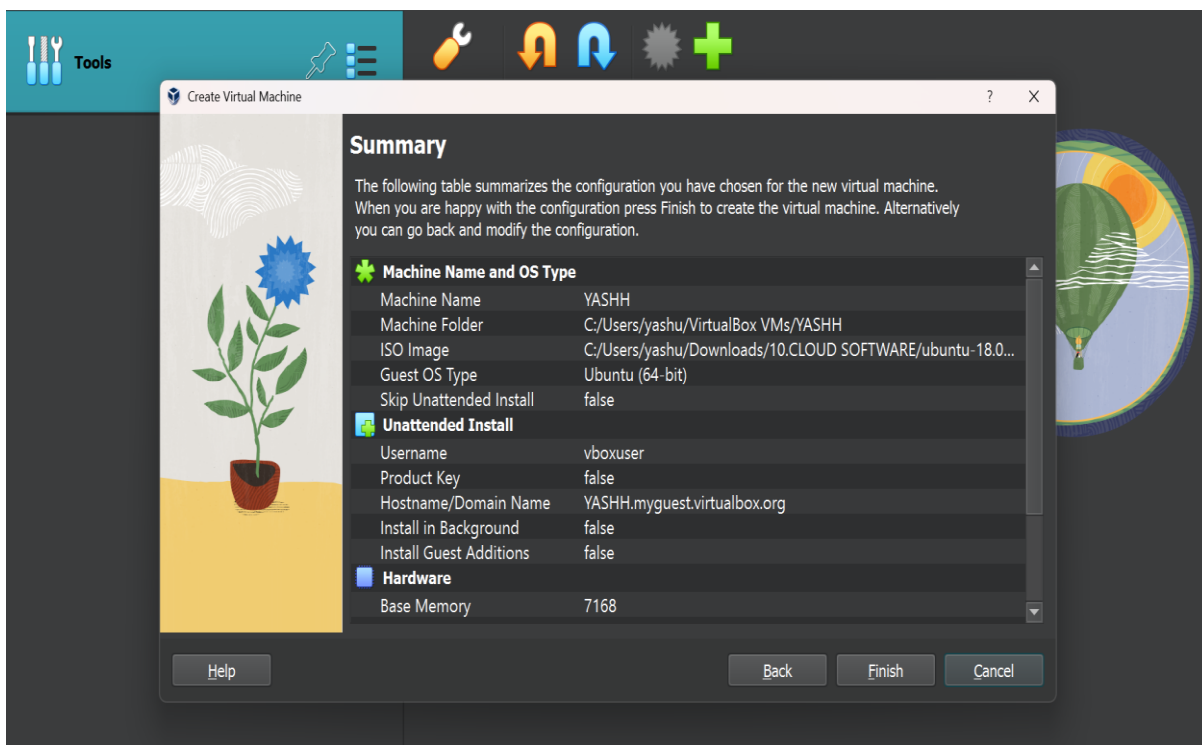
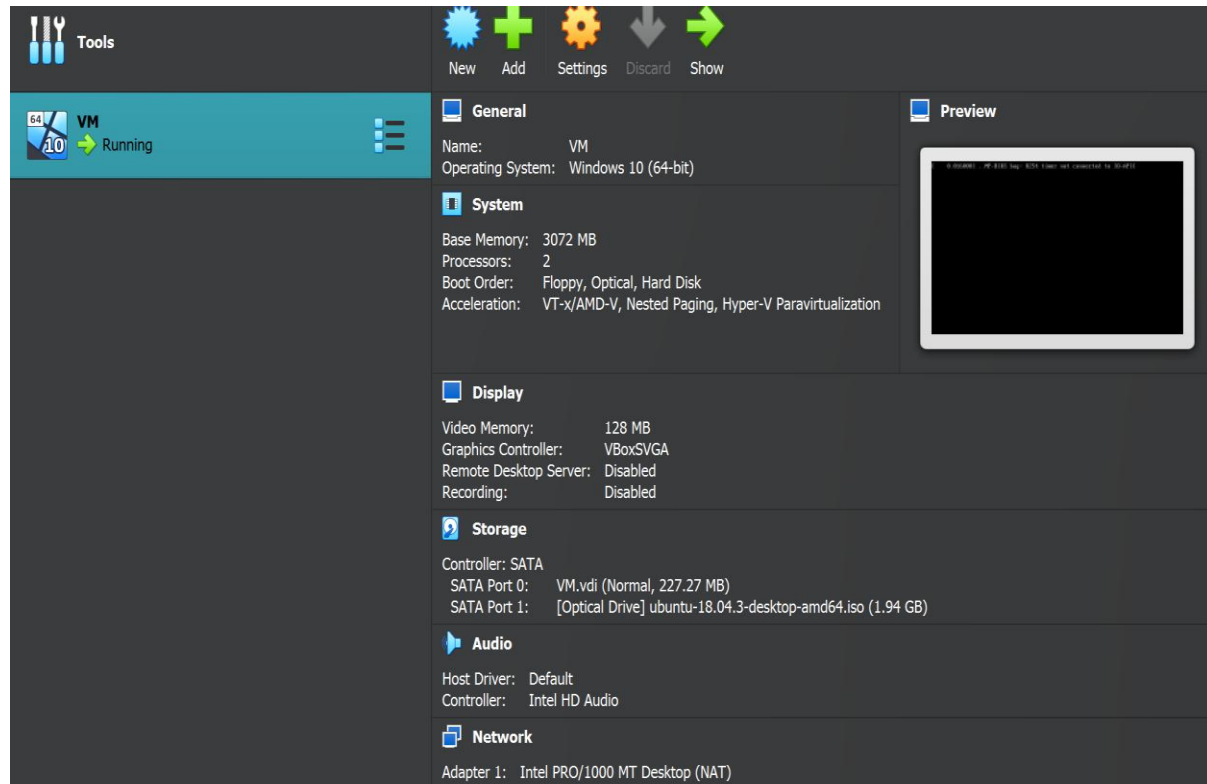


FIG : INSTALLATION OF VM IS FINISHED



Result: VirtualBox was successfully installed on Windows 10, and different Linux/Windows operating systems were installed as virtual machines using custom installation. The VMs were successfully configured and executed.

4.To write a procedure to run the virtual machine of different configuration and to check how many virtual machines can be utilized at a particular time.

Aim :

To run virtual machines with different configurations and determine the maximum number of virtual machines that can be utilized simultaneously on a host system.

Short Procedure :

1. Open the virtualization software such as VirtualBox.
2. Create a virtual machine with a selected configuration (CPU, RAM, and storage).
3. Install the required operating system in the virtual machine.
4. Start the virtual machine and verify that it is running properly.
5. Create additional virtual machines with different configurations.
6. Start multiple VMs simultaneously and monitor CPU, RAM, and storage utilization.
7. Gradually increase the number of running VMs until the host system reaches its resource limit.
8. Record the maximum number of VMs that can run simultaneously without significant performance degradation.
9. Compare the results for different VM configurations and note the resource utilization.

FIG : PROCESS OF CREATING A VM Like above two more VMs are installed

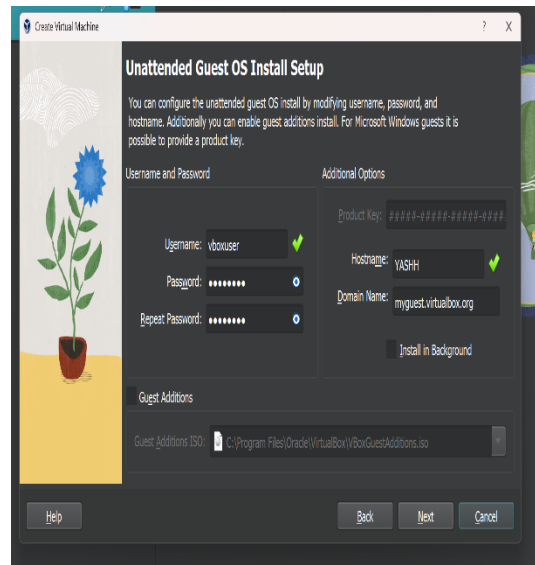
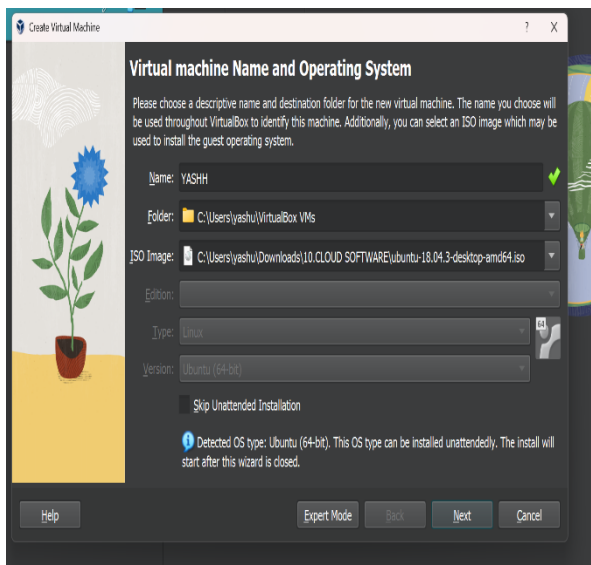


FIG : ACTIVITY STATUS OF VM YASHU

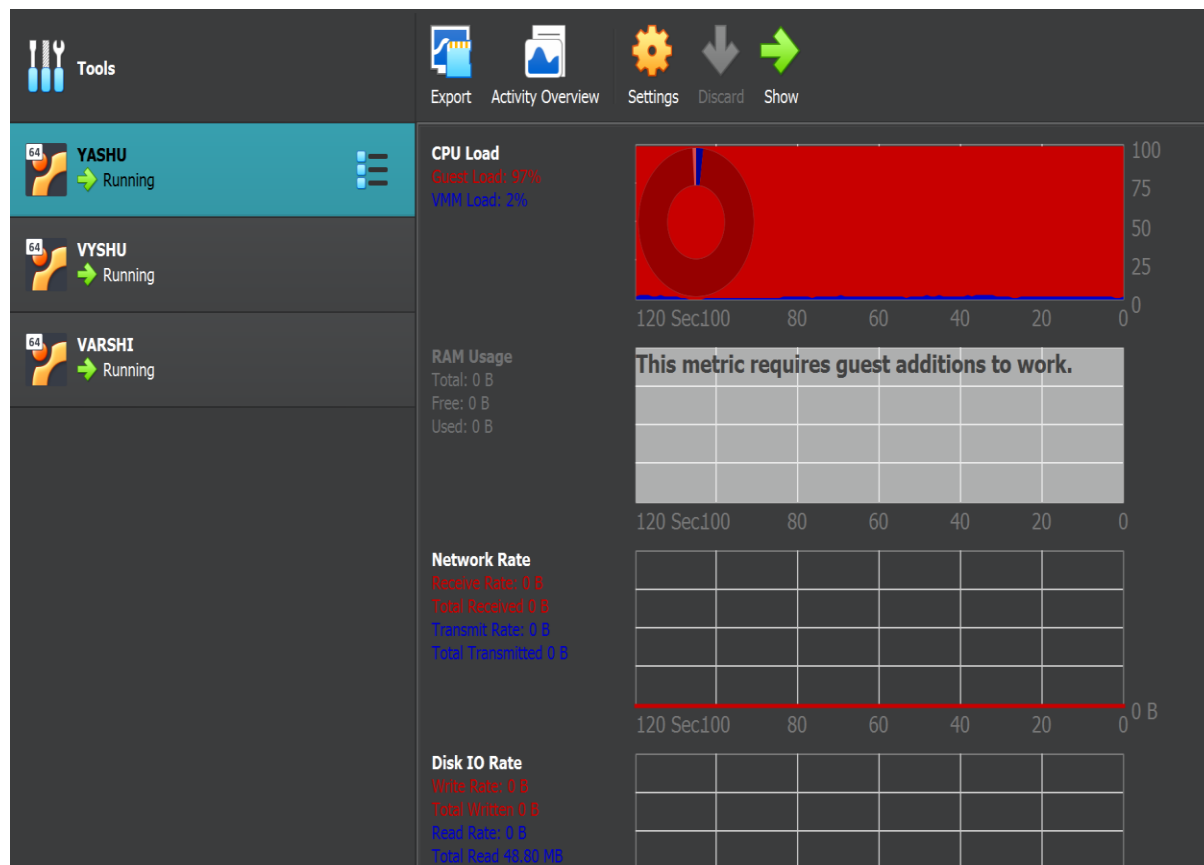


FIG : INSTALLATION OF VM VYSHU

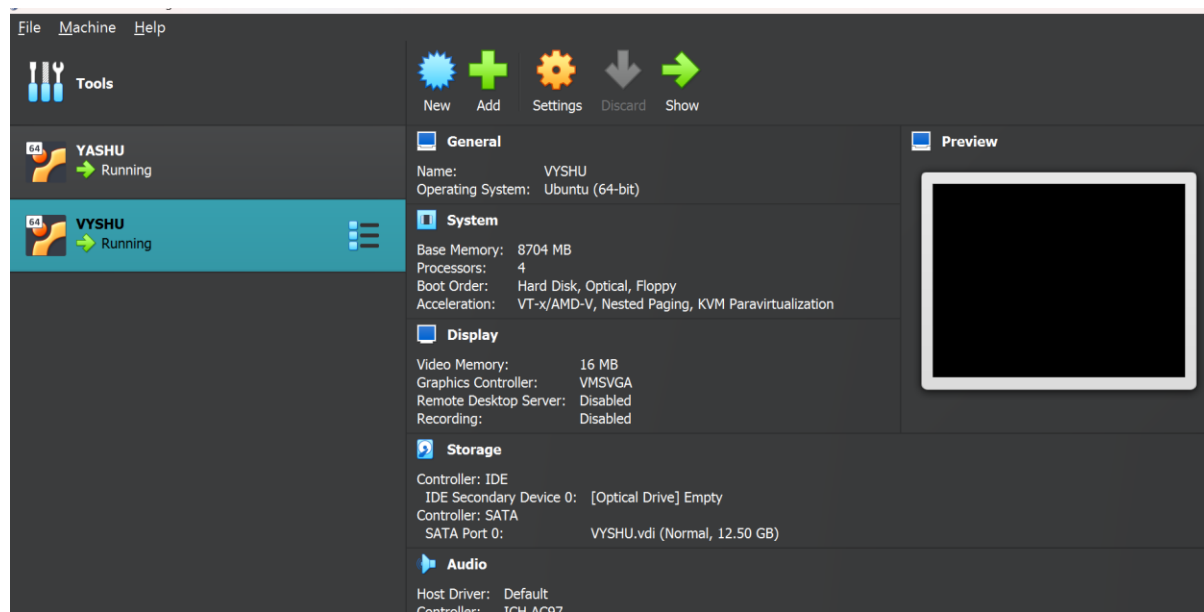


FIG : ACTIVITY STATUS OF VM VYSHU

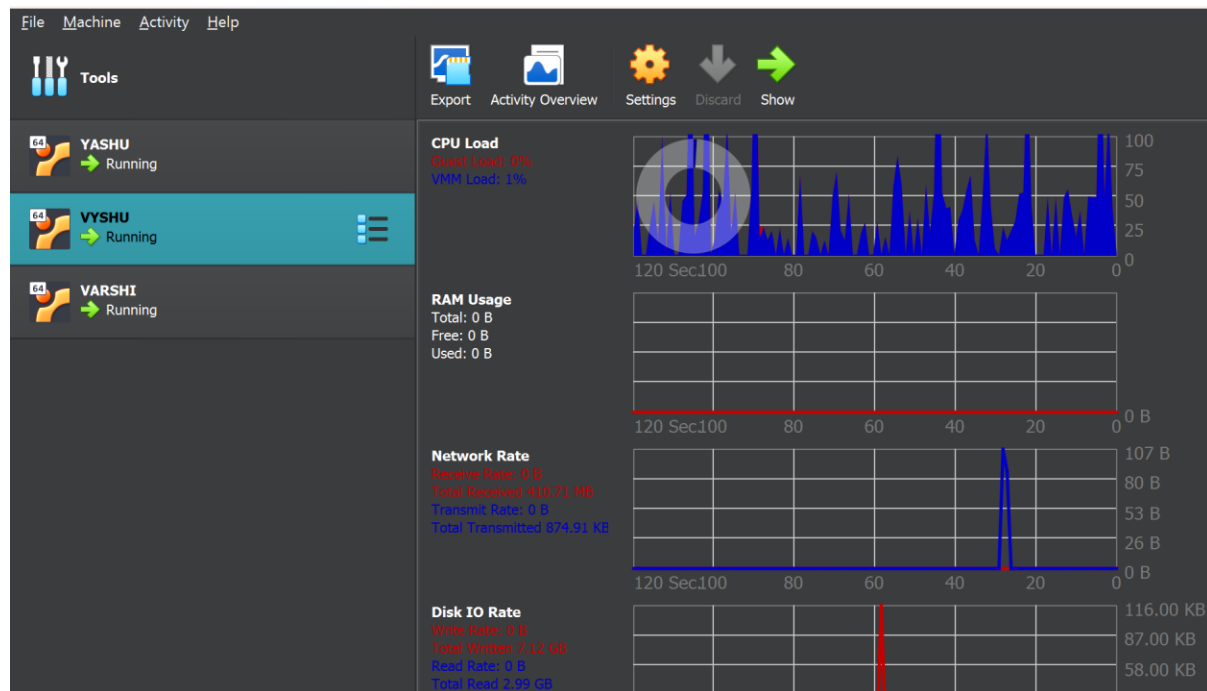


FIG : INSTALLATION OF VM VARSHI

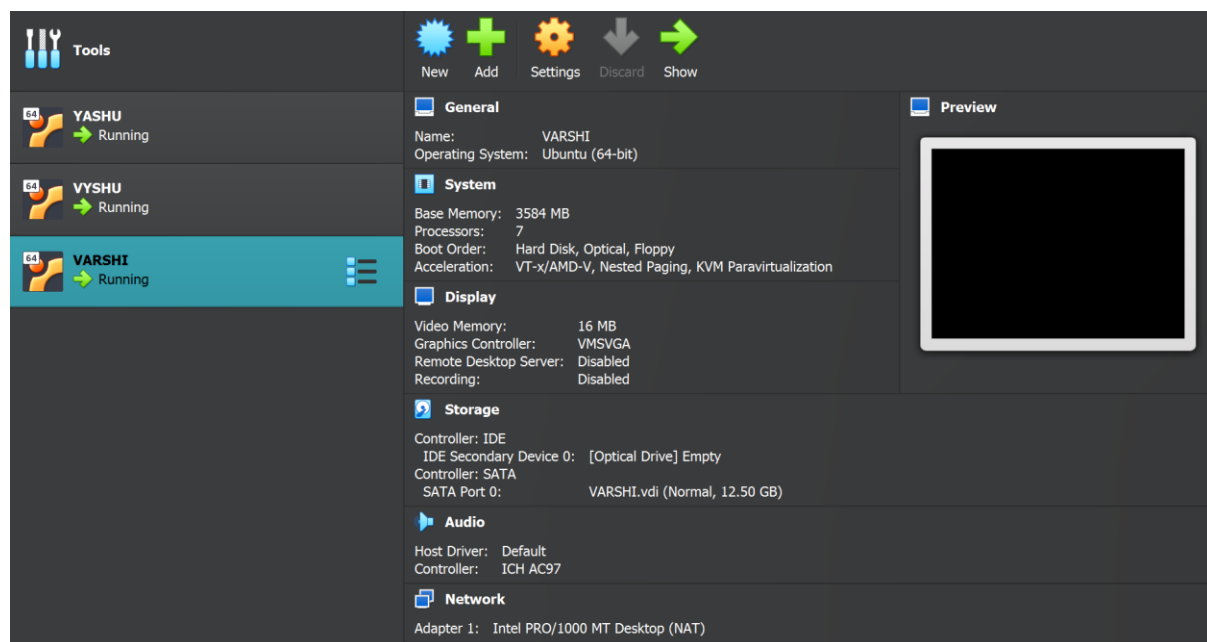
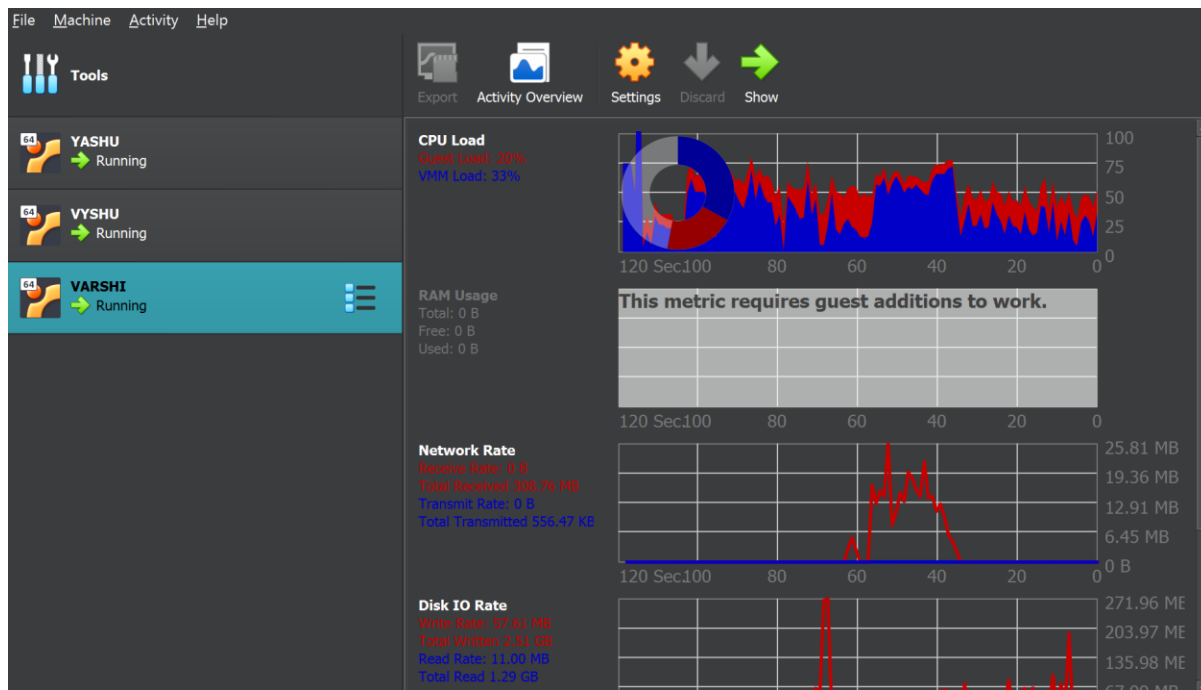


FIG : ACTIVITY STATUS OF VM VARSHI



Result: Virtual machines with different CPU, memory, and storage configurations were successfully created and executed simultaneously. The number of VMs that could be utilized at a particular time was determined by monitoring the host system's CPU, RAM, and disk utilization.